



COURSE SYLLABUS						
Course Code	ITEC 302	Course Title	Capstone Project 1		Course Credit	3
Course Description		Capstone Project 1 serves as the initial phase of the students’ culminating requirement in the BS Information Technology program. This course focuses on the conceptualization, research, and proposal development of an IT-based solution that addresses a real-world problem. Students will identify a research problem, conduct a comprehensive review of related literature, design appropriate methodologies, and prepare system analysis and design documentation. The course emphasizes critical thinking, collaboration, and adherence to research and ethical standards as students prepare for their proposal defense. By the end of the course, students are expected to produce and defend a complete Capstone Project proposal (Chapters 1–3).				
Pre-Requisites		COMP 015	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



**INSTITUTIONAL LEARNING
OUTCOMES
(ILO)**

As a polytechnic state university, PUP shall develop its students to possess:

1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
		Apply knowledge of computing, science, and mathematics appropriate to the discipline.								
		Analyze complex problems, and identify and define the computing requirements appropriate to its solution								
	Identify and analyze user needs and take them into account in the selection, creation,									



	evaluation and administration of computer-based systems.									
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.									



	Integrate IT-based solutions into the user environment effectively.									
	Assist in the creation of an effective IT project plan.									
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession									
	Understand best practices and standards and their applications.									
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.									
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.									
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.									
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development									

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Identify and define a real-world problem that can be addressed through an IT-based solution.								
	Conduct background research and needs analysis to establish project significance and feasibility.								
	Formulate clear project objectives, scope, and requirements aligned with user and organizational needs.								



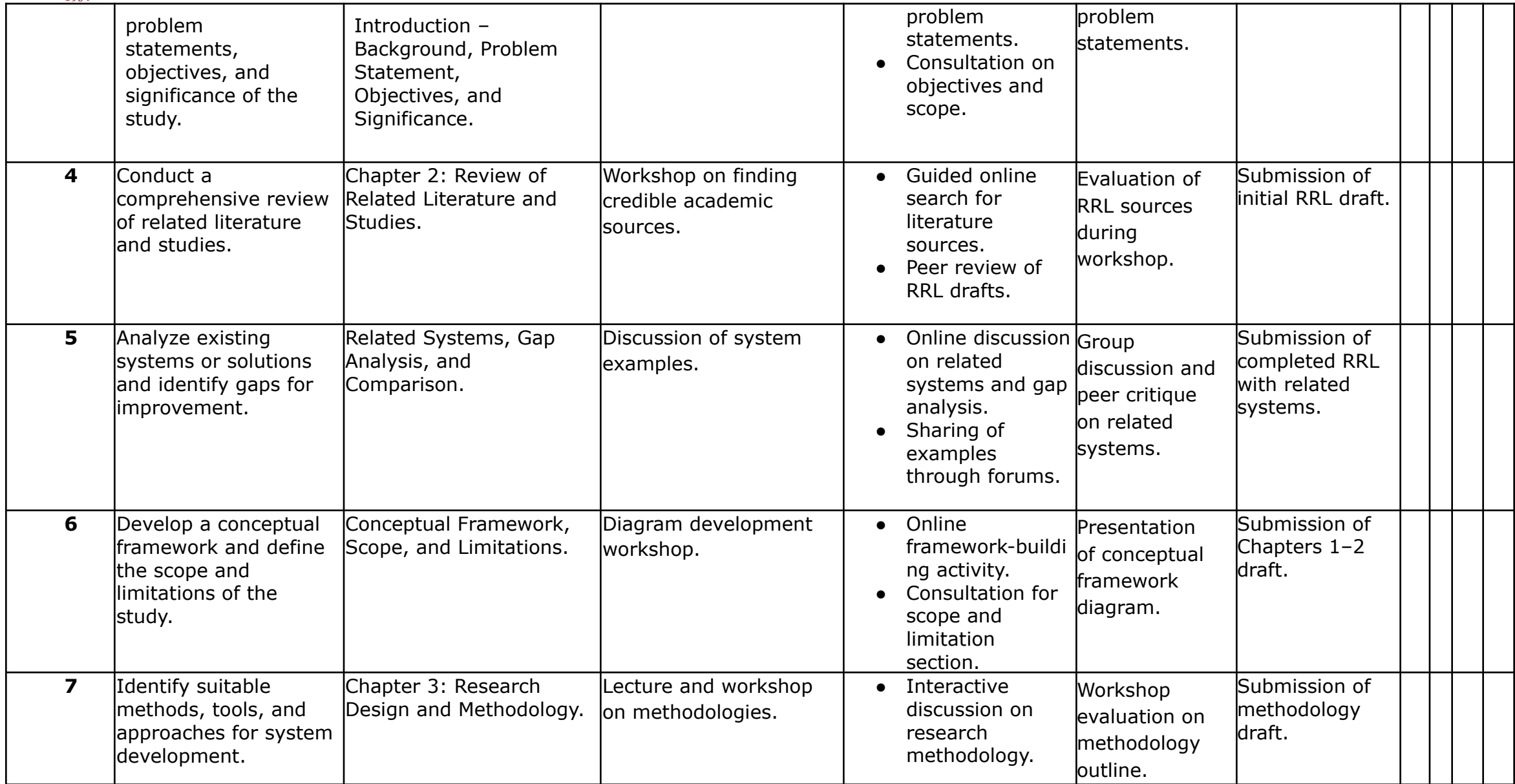
	Develop a comprehensive review of related literature and systems to support the proposed project.								
	Apply appropriate research methodologies and data-gathering techniques relevant to the project.								
	Design the system architecture, process models, and database schema for the proposed system.								
	Prepare a well-structured capstone project proposal document following institutional guidelines.								
	Present and defend the proposed project before a panel of evaluators with clarity and professionalism.								





OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignm ent to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	Understand the course objectives, requirements, and stages of Capstone Project 1; Identify researchable IT project problems aligned with institutional or community needs.	Course orientation and Capstone 1 overview; Identifying research problems and project opportunities; Group formation.	Orientation and brainstorming sessions; discussion of Capstone guidelines; topic consultation (synchronous and asynchronous).	- Online discussion of Capstone guidelines. - Brainstorming of project ideas. - Topic consultation via LMS.	Participation during orientation and consultations.	Submission of list of possible project topics.				
3	Formulate clear	Chapter 1:	Lecture and peer review.	Guided online writing of	Review and critique of	Submission of Chapter 1 draft.				

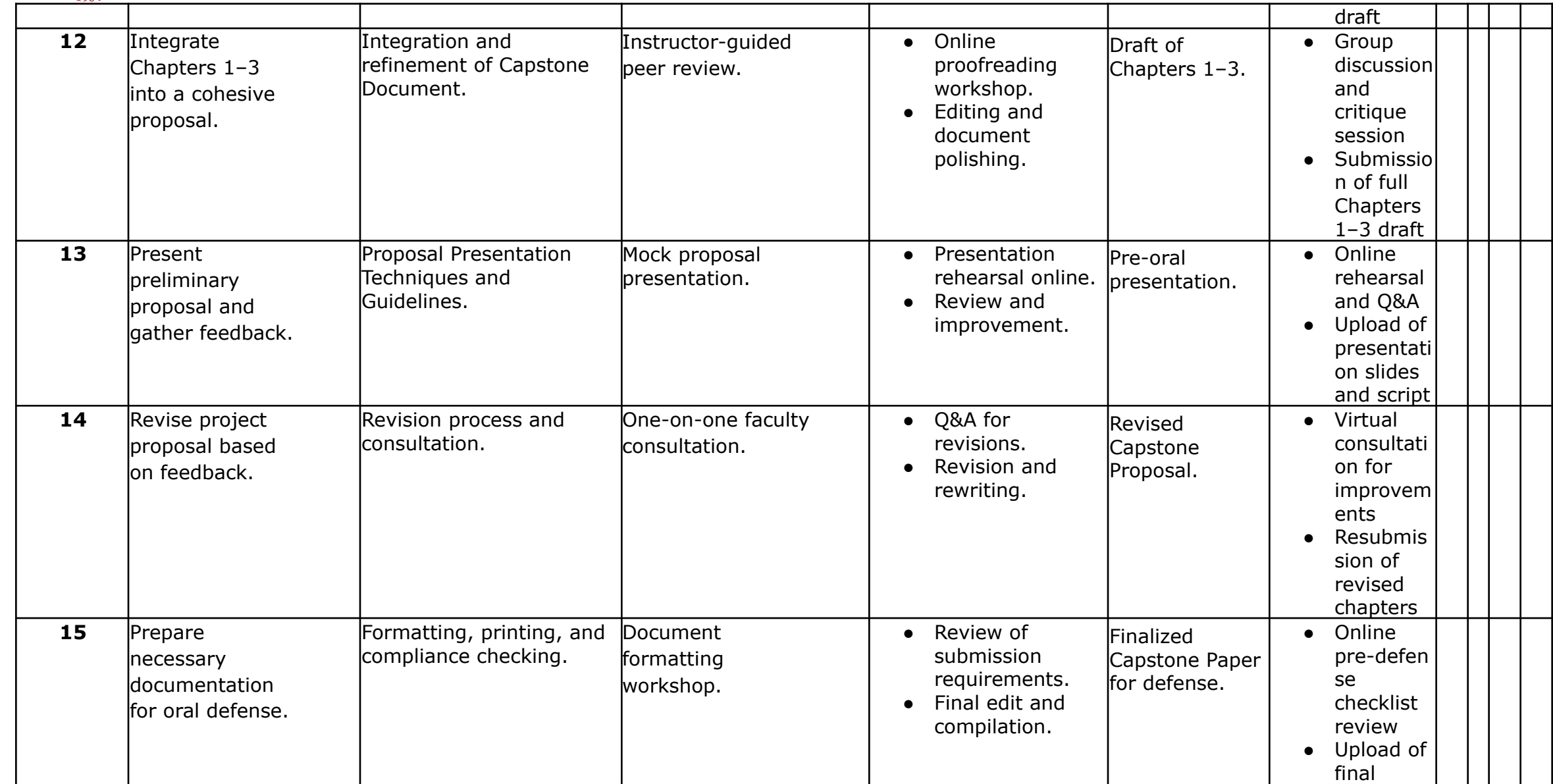


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				Online Q&A session for tool selection.					
8	Design and prepare data gathering instruments and define detailed system requirements based on user analysis.	Instruments, Data Collection, and Sampling Procedures; Requirements Analysis and Specification.	Workshop on survey/interview design and requirements gathering.	- Virtual workshop on survey/interview design. - Consultation for requirements documentation.	Checking of survey and interview tools.	Submission of validated instrument and requirements document.			
9	MIDTERM WRITTEN EXAMINATION								
10	Create initial system design models.	Use Case Diagram, DFD, and ERD.	Laboratory design session.	- Online demo of diagram tools. - Continued system modeling.	Draft of system models.	- Synchronous review of diagrams - Upload of system models file			
11	Prepare project timeline and development plan.	Operational, Technical, Economic, and Schedule Feasibility.	Discussion and case examples.	- Real-time consultation. - Feasibility writing and computation.	Feasibility study draft.	- Online feedback on feasibility computations - Upload of feasibility study			



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2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



STARS



- Employability
- Online Learning
- Internationalization



- Teaching
- Academic Development
- Inclusiveness



- Social Responsibility
- Sustainable Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)



THE WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI

Rank 39
STUDENT SUPPORT

Rank 50
SYMBOL/



**Times Higher Education
Impact Rankings 2024
DEBUTS AT 150+**



						formatted manuscript				
16	Deliver oral presentation effectively.	Proposal Defense proper.	Panel defense and Q&A.	• None (face-to-face defense week). • Review notes for improvement.	Oral defense and grading.	• Reflection post-defense via online form				
17	• Apply feedback and finalize approved project proposal. • Reflect on the capstone research process and future direction.	• Incorporating panel recommendations. • Reflection and documentation of learning.	• Consultation for revisions. • Reflection discussion and evaluation.	• Follow-up consultation online. • Revision and resubmission. • Closing synchronous reflection. • Closing synchronous reflection.	• Final approved manuscript. • Reflection paper / Course evaluation.	• Virtual consultation for final edits • Upload of approved final proposal • Online exit interview and reflection sharing • Submission of individual reflection paper and self-assessment				
18	FINAL EXAMINATION ()									



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
TAGUIG CAMPUS
Bachelor of Science in Information Technology
(BSIT)



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		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		• Quizzes					
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FINAL EXAMINATION

LIST OF REFERENCES

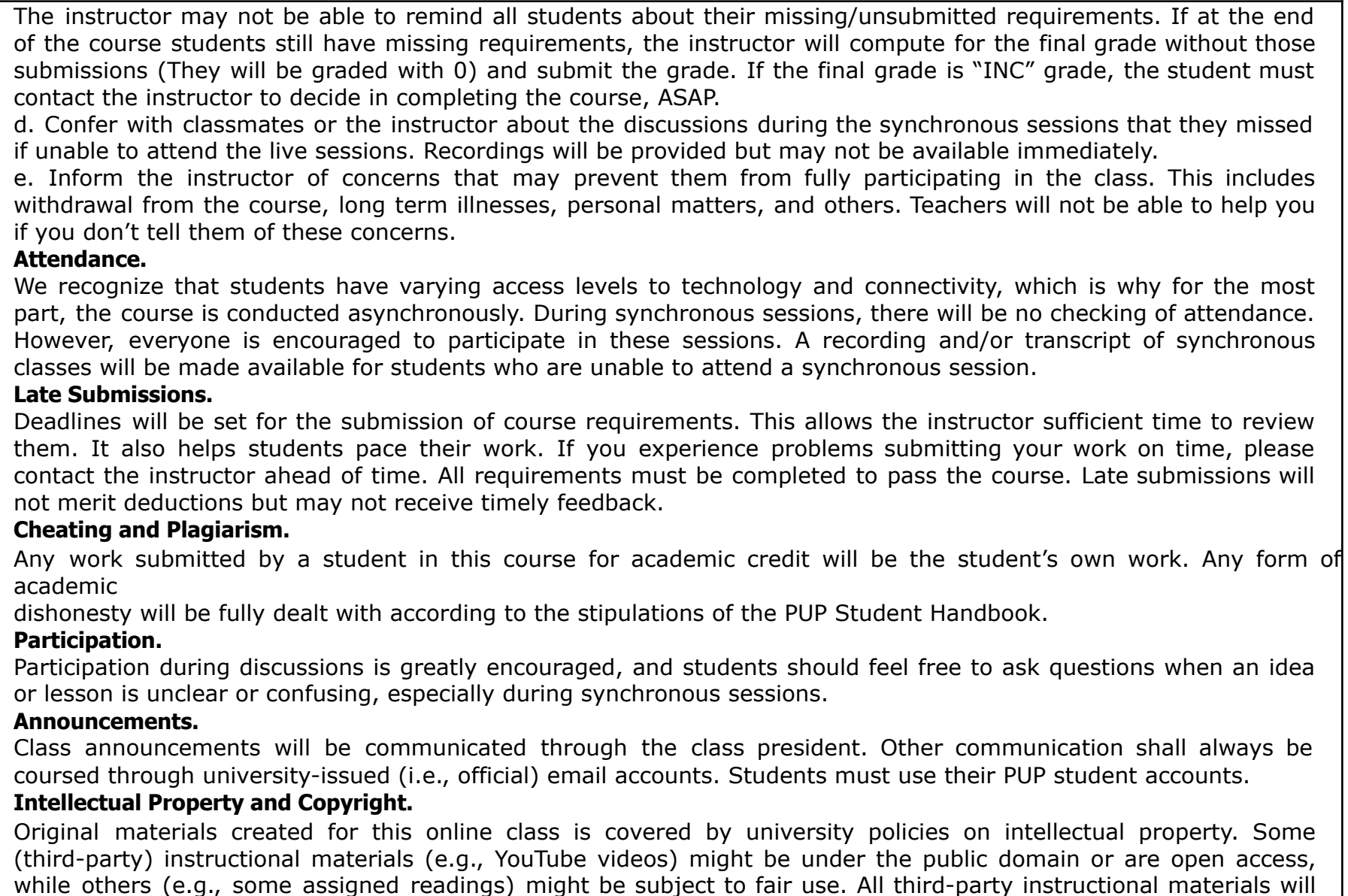
BOOKS:

WEBSITE:

- Lucidchart. (2023, November 8). How to create a conceptual framework for research. Lucidchart Blog.
<https://www.lucidchart.com/blog/how-to-make-a-conceptual-framework>
- Visual Paradigm. (2023, May 12). How to create data flow diagrams (DFD): A beginner’s guide. Visual Paradigm.
<https://www.visual-paradigm.com/guide/data-flow-diagram/what-is-data-flow-diagram/>
- IBM. (2024, February 15). What is system analysis and design? IBM Developer. <https://developer.ibm.com/articles/system-analysis-design/>
- Project Management Institute. (2023, October 2). What is a feasibility study and why is it important? PMI Knowledge Center.
<https://www.pmi.org/learning/library/feasibility-study-importance-12345>
- ResearchGate. (2024, January 9). Best practices in writing a research methodology for IT capstone projects. ResearchGate.
<https://www.researchgate.net/publication/IT-capstone-methodology-best-practices>



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.





	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.																																	
COURSE REQUIREMENT/S	GRADING SYSTEM																																	
Students are required to prepare and submit a comprehensive Capstone Project Proposal covering Chapters 1 to 3. They must develop system analysis and design artifacts, including use case diagrams, data flow diagrams, entity-relationship diagrams, and a conceptual framework. A feasibility study assessing the project’s technical, operational, and economic viability must be completed. Students are expected to attend regular consultations and submit progress reports to demonstrate consistent development. They must also present and defend their proposal before a panel and submit a final revised and approved version of their project proposal.	<table><tr><th colspan="2">Mid Term</th><th colspan="2">Final Term</th></tr><tr><td>Class Standing</td><td>70%</td><td>Class Standing</td><td>70%</td></tr><tr><td>Exercises/Activities</td><td></td><td>Exercises/Activities</td><td></td></tr><tr><td>Summative Quizzes</td><td></td><td>Summative</td><td></td></tr><tr><td>Recitation and Attendance</td><td></td><td>Quizzes Recitation and Attendance</td><td></td></tr><tr><td>Reflection Paper</td><td></td><td>Reflection Paper</td><td></td></tr><tr><td>Midterm Examinations</td><td><u>30%</u></td><td>Final Examinations</td><td>30%</td></tr><tr><td></td><td>100%</td><td></td><td>100%</td></tr></table>	Mid Term		Final Term		Class Standing	70%	Class Standing	70%	Exercises/Activities		Exercises/Activities		Summative Quizzes		Summative		Recitation and Attendance		Quizzes Recitation and Attendance		Reflection Paper		Reflection Paper		Midterm Examinations	<u>30%</u>	Final Examinations	30%		100%		100%	
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Final Grade = (Midterm Grade + Final Term Grade) / 2																																		



School Year: 2526	Semester: 2nd Semester	E-Mail Address:
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Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
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